INCH-POUND

MIL-DTL-13777/6C
20 July 2011
SUPERSEDING
MIL-C-13777/6B
1 September 1966

DETAIL SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL
60, 65, AND 78 CONDUCTORS

Inactive for new design after 1 June 1998.

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-13777.

REQUIREMENTS:

Dimensions and configuration: See applicable figure and design data for the following cable types specified in table I. Design data is specified in table II.

TABLE I. Cable types.

601330 601495S	601665S 601734S	601845S 601931S	651125 782166S

TABLE II. Design data.

	Design data				
Type designation	601330		601495S		601931S
Figure number	1		2		1
Total wires	60		60		60
Number of conductors & AWG #	60/#18	51/#18		9/#18S	60/#18S
Insulation					
Minimum average thickness	0.015”		0.015”		0.015”
Spark test voltage	3000		3000		3000
Inspection test voltage	1500		1500		1500
Cabling					
Layer 1	Filler		Filler		Filler
Layer 2					
(a) Number of wires	6		6		6
(b) AWG #	#18		#18		#18S
(c) Maximum lay					
Layer 3					
(a) Number of wires	12		9		12
(b) AWG #	#18		#18S		#18S
(c) Maximum lay					
Layer 4					
(a) Number of wires	18		20		18
(b) AWG #	#18		#18		#18S
(c) Maximum lay					
Layer 5					
(a) Number of wires	24		25		24
(b) AWG #	#18		#18		#18S
(c) Maximum lay	9.00”		12.00”		10.00”
Sheath					
Number of layers	2		2		2
Total thickness minimum	0.141”		0.141”		0.188”
Minimum OD cable	1.300”		1.470”		1.881”
Maximum OD cable	1.360”		1.520”		1.981”

See notes at end of table.

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TABLE II. Design data –Continued.

	Design data			
Type designation	601665S[1/,2/]	601734S[3/,4/,5/]	601845S[6/,7/,8/,9/]	651125
Figure number	5	4	3	6
Total wires	60	60	60	65
Number of conductors & AWG #				65/#20
Insulation				
Minimum average thickness	0.015”	0.015”	0.015”	0.015”
Spark test voltage	3000	3000	3000	3000
Inspection test voltage	1500	1500	1500	1500
Cabling				
Layer 1	Filler	Filler	Filler	Filler in center
(a) Number of wires				8
(b) AWG #				#20
(c) Maximum lay				
Layer 2				
(a) Number of wires				13
(b) AWG #				#20
(c) Maximum lay				
Layer 3				
(a) Number of wires	15/#18S	20		19
(b) AWG #				#20
(c) Maximum lay				
Layer 4				
(a) Number of wires	31/#18			25
(b) AWG #				#20
(c) Maximum lay	11.00”	11.50”	13.00”	
Sheath				
Number of layers	2	2	2	2
Total thickness minimum	0.156”	0.156”	0.109”	0.098”
Minimum OD cable	1.615”	1.684”	1.820”	1.100”
Maximum OD cable	1.715”	1.784”	1.870”	1.150”

See notes at end of table.

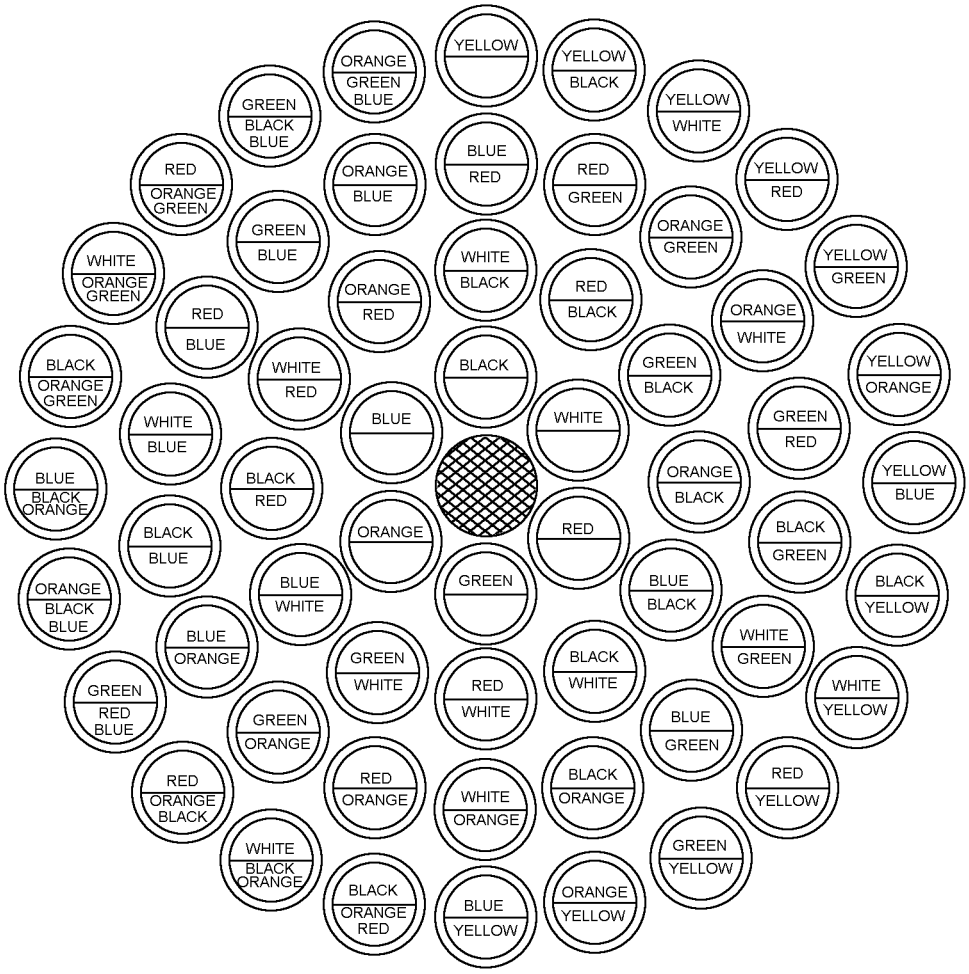
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TABLE II. Design data – Continued.

Design data						
Type designation			782166S			
Figure number			7			
Total wires			78			
Number of conductors & AWG #	64/#20	4/#20S	2/#14		2/#8	6/#4
Insulation			0.018"			
Minimum average thickness	0.015"				0.025"	0.063"
Spark test voltage			3000			
Inspection test voltage	1500		1800		2500	3500
Cabling						
Minimum average thickness						
Layer 1			Filler			
Layer 2						
(a) Number of wires	2	6	12			
(b) AWG #	#14	#4				4/#20S
(c) Maximum lay			8			
Layer 3						
(a) Number of wires			66			
(b) AWG #	2/#8					64/#20
(c) Maximum lay			12.00"			
Sheath						
Number of layers			2			
Total thickness minimum			0.183"			
Minimum OD cable			2.116"			
Maximum OD cable			2.216"			
Maximum weight (lbs./1000ft.)			2770			

1/ Number of conductors and AWG# for 601665S: 35/#18, 5 Pairs, Twisted & Shielded, 15/#18S.
2/ Number of wires for layer 2 of 601665S: 5 Pairs, Twisted, & Shielded, 4/#18.
3/ Number of conductors and AWG# for 601734S: 20/#18, 20 Pairs, Twisted, & Shielded.
4/ Number of wires layer 2 for 601734S: 6 Pairs. Twisted, & Shielded.
5/ Number of wires layer 4 for 601734S: 14 Pairs. Twisted, & Shielded.
6/ Number of conductors and AWG# for 601845S: 20 Pairs, Twisted & Shielded, 20 Single Shielded.
7/ Number of wires for layer 2 of 601845S: 5 Pairs, Twisted & Shielded, 3 Singles Shielded.
8/ Number of wires for layer 3 of 601845S: 17 Shielded.
9/ Number of wires for layer 4 for 601845S: 15 Pairs, Twisted, & Shielded.

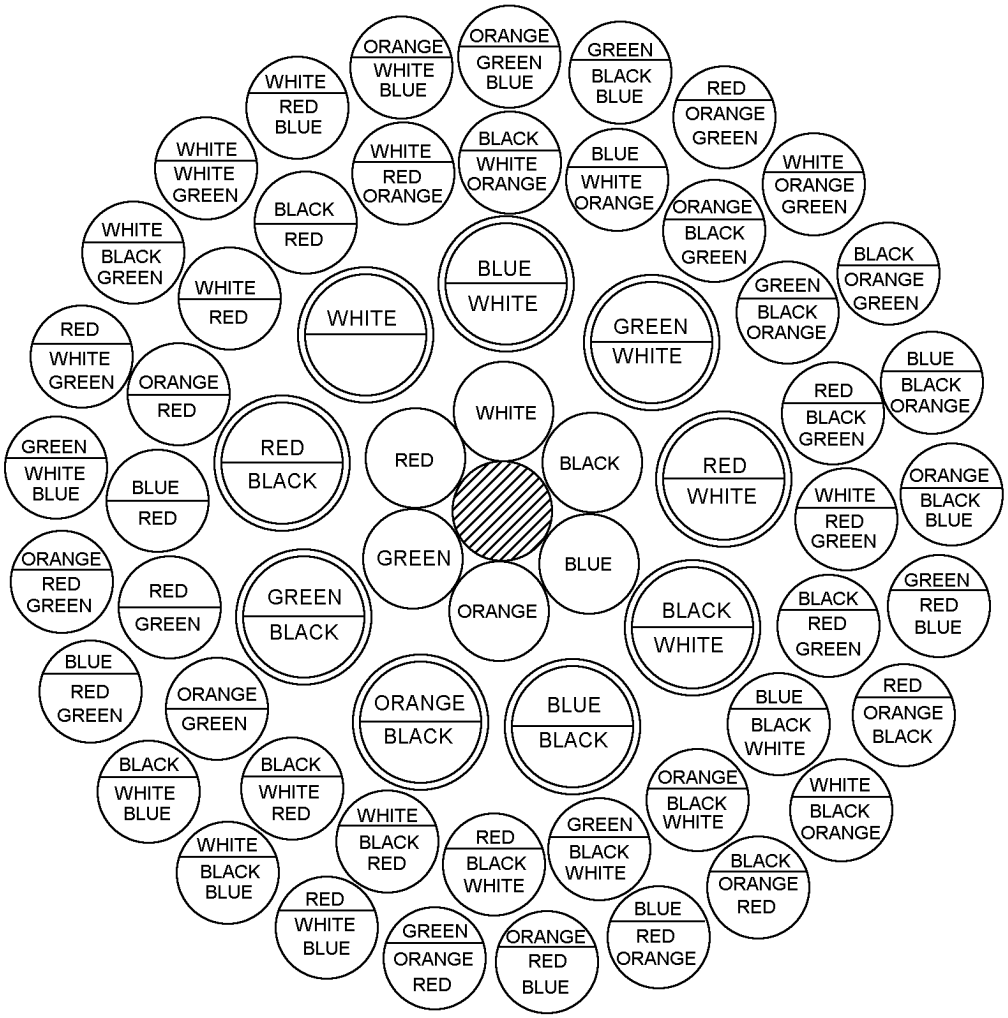
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NOTE: Double circle indicates shielded conductors.

FIGURE 1. Cable types (601330, 601931S).

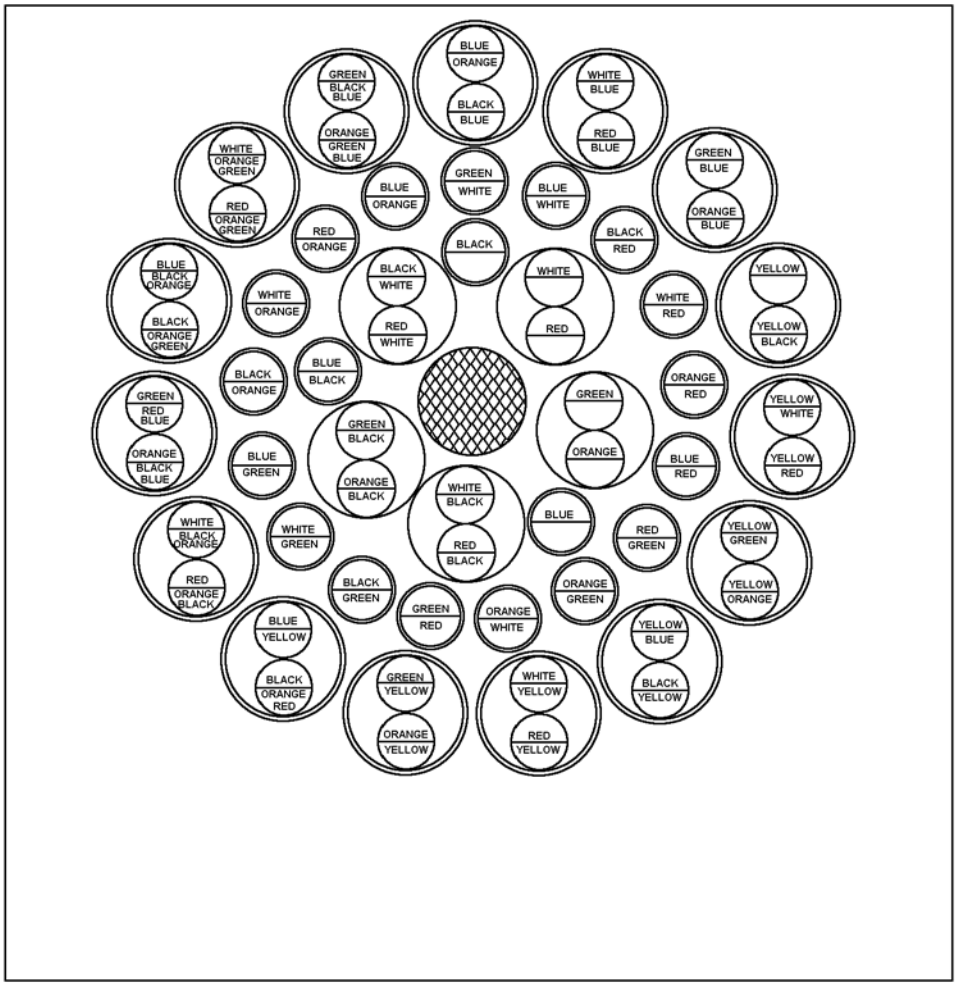
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NOTE: Double circle indicates shielded conductors.

FIGURE 2. Cable type (6014958).

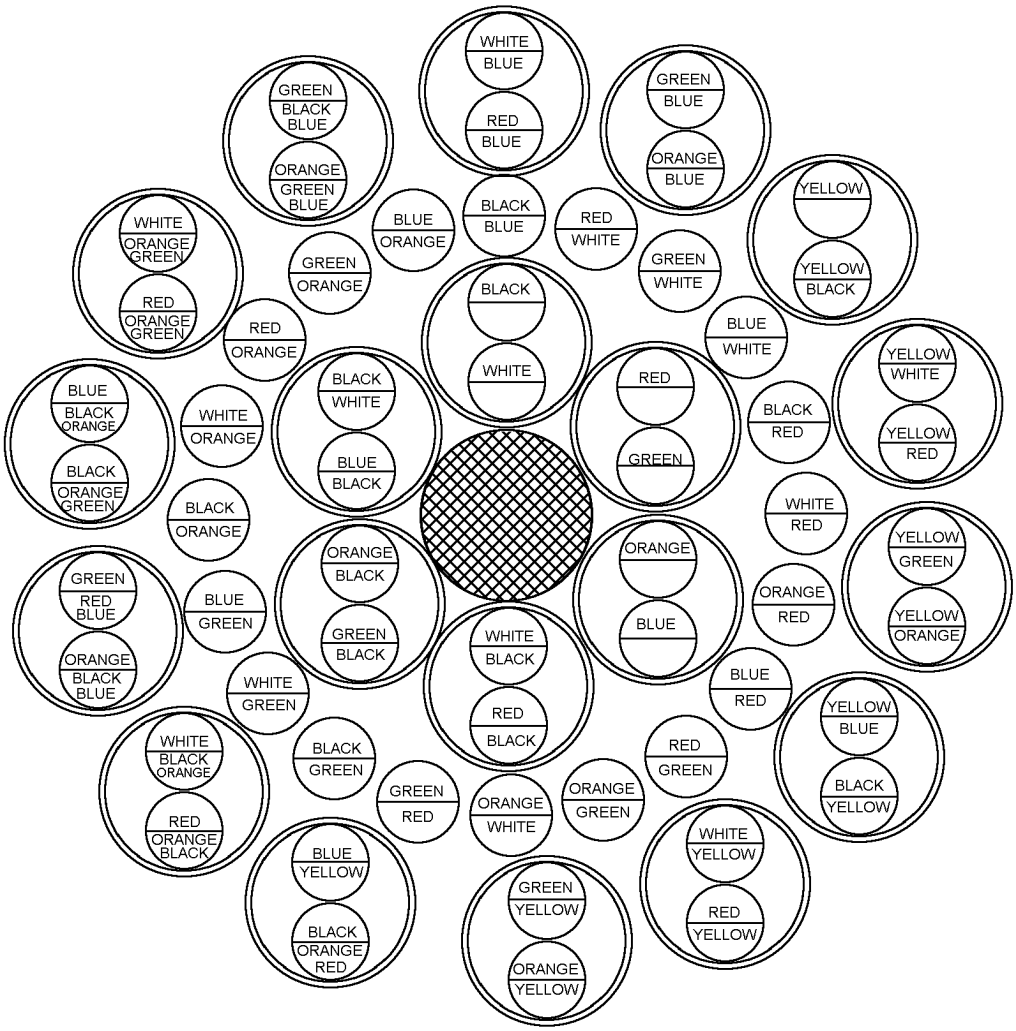
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NOTE: Double circle indicates shielded conductors.

FIGURE 3. Cable type (601845S).

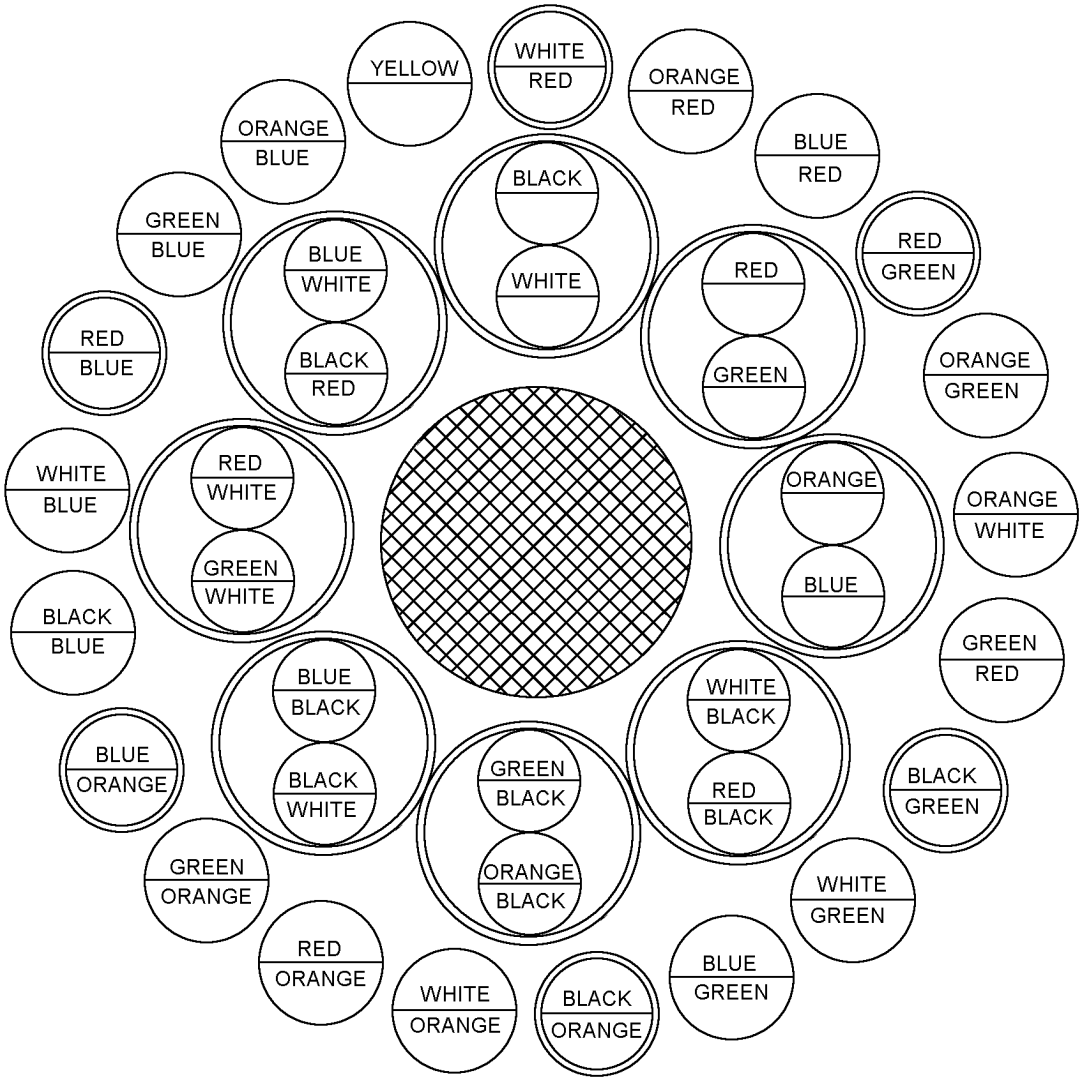
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NOTE: Double circle indicates shielded conductors.

FIGURE 4. Cable type (6017348).

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NOTE: Double circle indicates shielded conductors.

FIGURE 5. Cable type (601665S).

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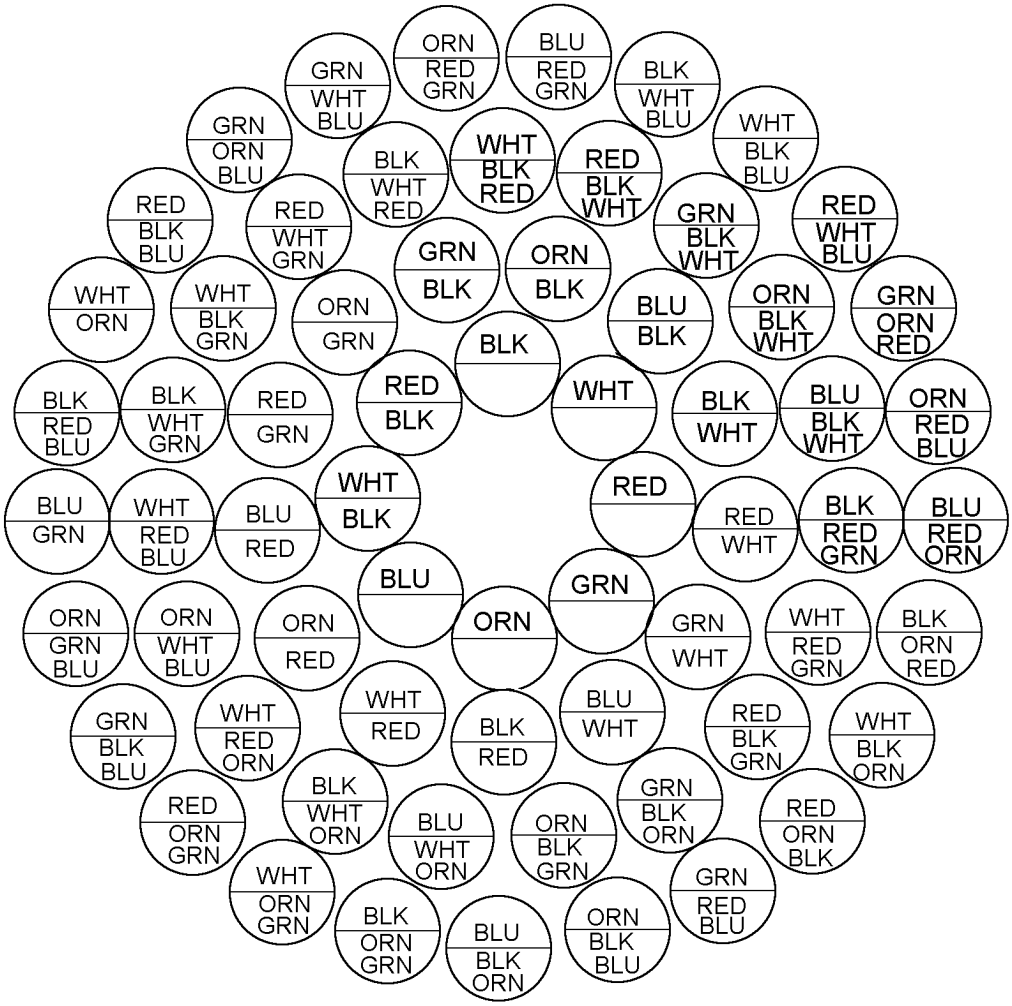
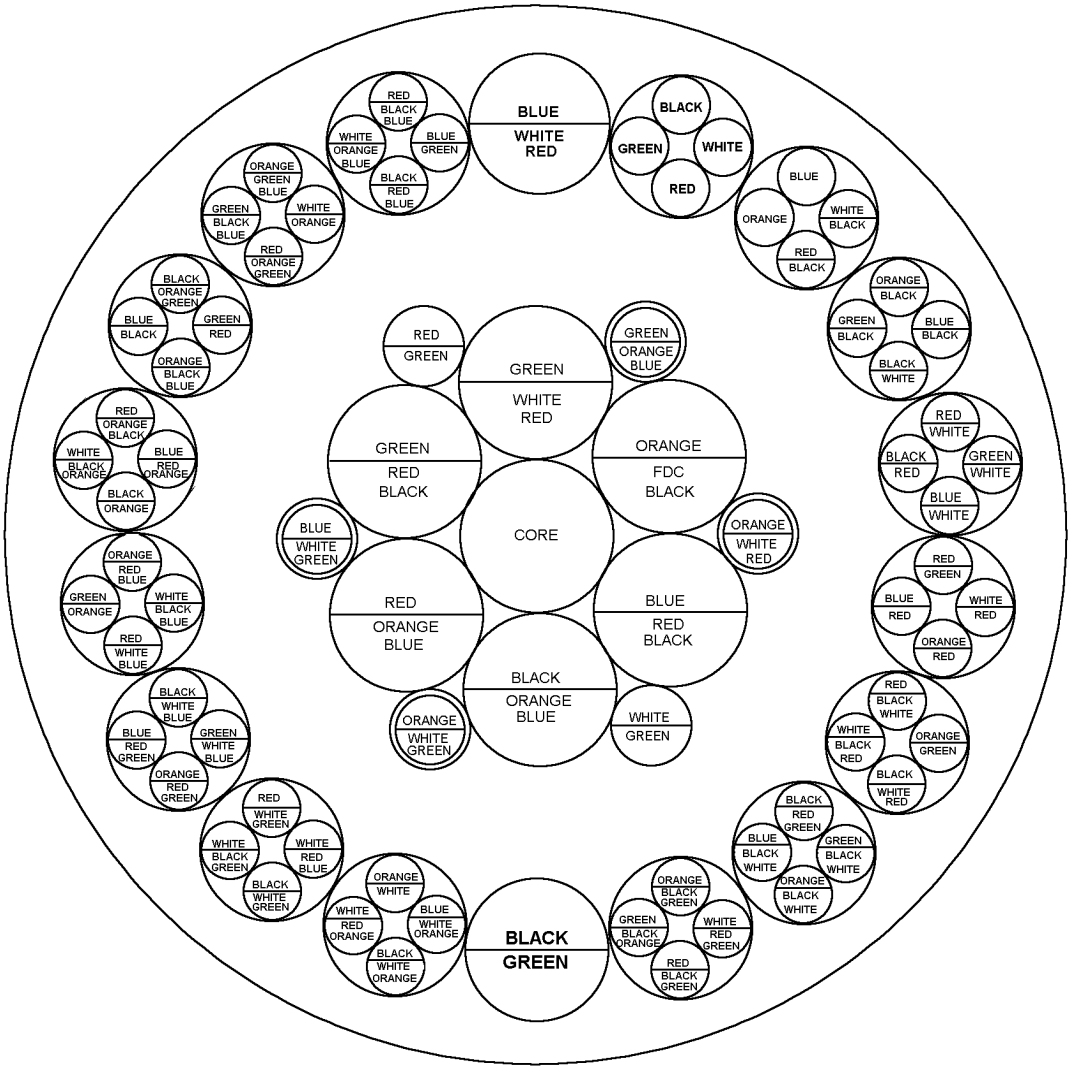


FIGURE 6. Cable type (651125).

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NOTE: Double circle indicates shielded conductors.

FIGURE 7. Cable type (782166S).

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Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issues due to the extensiveness of the changes.

Referenced documents. This document references MIL-DTL-13777.

CONCLUDING MATERIAL

Custodians:
Army – AR
Navy – SH
Air Force – 85
DLA – CC

Preparing activity:
DLA - CC

(Project 6145-2011-009)

Review activities:
Army – AT, CR, CR4, MI
Navy – MC

Note: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information using the Assist Online database at https://assist.daps.dla.mil.

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